

Second Year Engineering Course Content

Comprehensive Syllabi for Fluid Mechanics, Circuit Analysis, Thermodynamics, and Beam Analysis

1. Fluid Mechanics

ENG 201

Course Overview: Introduction to the behavior of fluids at rest (statics) and in motion (dynamics), focusing on fundamental governing equations, dimensional analysis, and engineering applications.

Module 1: Fundamental Concepts & Properties

- Definition of a fluid and continuum hypothesis
- Density, specific weight, specific gravity, dynamic/kinematic viscosity
- Surface tension, capillarity, and compressibility

Module 3: Kinematics of Fluid Flow

- Velocity fields, streamlines, streaklines, and pathlines
- Steady/unsteady, laminar/turbulent, compressible/incompressible flows
- Continuity equation in integral and differential forms

Module 5: Dimensional Analysis & Similitude

- Buckingham Π Theorem and dimensionless parameters
- Primary groups: Reynolds, Froude, and Euler numbers
- Model testing, dynamic similarity, and scaling laws

Module 2: Fluid Statics

- Pressure distribution and hydrostatic forces on submerged surfaces
- Manometry and pressure measurement techniques
- Buoyancy, Archimedes' principle, and stability of floating bodies

Module 4: Fluid Dynamics & Conservation Laws

- Euler's equation of motion along a streamline
- Derivation/applications of Bernoulli's equation (Venturi, Pitot tube)
- Linear momentum equation and forces on pipe bends and vanes

Module 6: Viscous Internal Flow (Pipe Flow)

- Laminar flow in circular pipes (Hagen-Poiseuille equation)
- Darcy-Weisbach friction factor and Moody diagram
- Minor head losses in fittings, valves, and pipe networks

2. Circuit Analysis

ENG 202

Course Overview: Comprehensive analysis of direct current (DC) and alternating current (AC) linear networks, transient response, operational amplifiers, and frequency-domain techniques.

Module 1: Basic Circuit Laws & Analysis

- Lumped circuit elements: resistors, inductors, capacitors, sources
- Ohm's Law, Kirchhoff's Current (KCL) and Voltage (KVL) Laws
- Nodal analysis and Mesh current analysis techniques

Module 3: Operational Amplifiers (Op-Amps)

- Ideal op-amp characteristics and golden rules
- Inverting, non-inverting, summing, and difference configurations
- Integrators, differentiators, and practical non-idealities

Module 5: AC Steady-State Analysis

- Sinusoids, phasors, and complex impedance/admittance
- AC mesh/node analysis and network theorems
- Instantaneous, average, reactive, complex power, power factor

Module 2: Circuit Theorems

- Linearity and Superposition theorem
- Thévenin's and Norton's equivalent circuits
- Maximum power transfer theorem and source transformations

Module 4: Transient Circuit Analysis

- Natural and step response of first-order **RC** and **RL** circuits
- Step response of series and parallel second-order **RLC** circuits
- Overdamped, critically damped, and underdamped conditions

Module 6: Two-Port Networks & Laplace Domain

- Frequency response, series/parallel resonance, passive filters
- Laplace transform application to circuit analysis (**s**-domain)
- Two-port parameters: **Z**, **Y**, **h**, and **ABCD** matrices

3. Thermodynamics

ENG 203

Course Overview: Fundamental laws governing heat, work, energy transformations, properties of pure substances, and thermodynamic power/refrigeration cycles.

Module 1: Fundamental Concepts & Zeroth Law

- System, surroundings, boundaries, state, and equilibrium
- Temperature, Zeroth Law of Thermodynamics
- Work and heat definitions, sign conventions, path vs. state functions

Module 3: First Law of Thermodynamics

- Closed systems: Energy conservation, boundary work ($\int P dv$)
- Open systems: Steady-Flow Energy Equation (SFEE)
- Applications to nozzles, turbines, compressors, heat exchangers

Module 5: Entropy & Exergy

- Clausius inequality and principle of entropy increase
- Entropy change for ideal gases, liquids, and solids
- Isentropic efficiencies; exergy (availability) and second-law efficiency

Module 2: Properties of Pure Substances

- Phase-change processes, P - v - T surface and phase diagrams
- Ideal gas law, compressibility factor (Z), real gas equations
- Property tables: internal energy (u), enthalpy (h), entropy (s)

Module 4: Second Law & Carnot Engine

- Kelvin-Planck and Clausius statements; heat engines and pumps
- Reversibility, Carnot cycle, and Carnot principles
- Absolute thermodynamic temperature scale

Module 6: Power & Refrigeration Cycles

- Air-standard gas cycles: Otto, Diesel, and Brayton cycles
- Vapor power cycles: Simple Rankine cycle
- Vapor-compression refrigeration cycle

4. Beam Analysis

ENG 204

Course Overview: Mechanics and structural analysis of internal forces, bending/shear stresses, elastic deflections, and stability in structural beams under static loads.

Module 1: Determinate Structural Systems

- Beam types: simply supported, cantilever, overhanging, continuous
- Reaction forces under point, distributed (w), and moment loads
- Static determinacy and degree of indeterminacy

Module 3: Bending Stresses in Beams

- Euler-Bernoulli beam theory assumptions
- Flexure formula derivation: $\sigma = -My / I$
- Section modulus (S) and stress profiles (I, T, circular sections)

Module 5: Beam Deflection (Elastic Curve)

- Differential equation: $EI (d^2w / dx^2) = M(x)$
- Double Integration method and Macaulay's method
- Moment-Area theorems and Conjugate Beam method

Module 2: Internal Shear & Bending Moment

- Relationships between load, shear force (V), and moment (M)
- Shear Force Diagrams (SFD) & Bending Moment Diagrams (BMD)
- Maximum shear/moment locations and points of contraflexure

Module 4: Shear Stresses in Beams

- Transverse shear stress formula: $\tau = VQ / Ib$
- Shear stress distribution across standard cross-sections
- Shear flow and shear center concepts for unsymmetrical sections

Module 6: Indeterminate Beams & Buckling

- Method of Superposition for propped cantilevers and fixed beams
- Theorem of Three Moments (Clapeyron's equation)
- Euler's critical load formula for column buckling (P_{cr})

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